AI disclosure

I saw the note on my hw4, and I am extremely sorry for not including this.

For hw5 and previous assignments, I used Claude to learn and explore very direct topics/ideas (a succinct intuitive understanding, then I ask for code example implementations to get ideas and be able to connect the intuition to structured steps). Also, claude code has access to my syllabus file and class files and it won't produce any hw solutions directly for me even if I try.

for hw5 specifically I used Claude for these topics and questions:

"open up hw5 and list all the major topics I should understand; be succinct"

" for each of these, I need a short intuitive explanation, a runnable python code use made up data, steps for how I should apply this to hw5- check and read what the subparts are; keep explanations concise!!"

some more questions within these topics:

1. Decision trees and diagnosis labels: How do I represent two yes/no diagnoses as 4 combined classes, fit and interpret a decision tree, and convert its paths into if-then rules?
2. Tree pruning: how does cost complexity pruning work? What is ccp_alpha, and how do I explain choosing a simpler tree when the full tree has better cross-val accuracy?
3. Preprocessing and cross-validation: how do I preserve the first 1495 row training split, remove identifiers, and handle missing values? Why does imputation and scaling happen inside each cross-val fold, and whats with the pipeline in the context of all this??
4. OLS, Ridge, and LASSO: quickly run me through their objectives again? Explain test mse, choosing regularization strength through cross-validation, why scaling matters, and how to compare standardized versus unstandardized lasso and extract selected variables. Try to do this in the context of econ 513 terms that we use- check ECON 513 folder on desktop.
5. Principal component regression: What does PCA do intuitively, why scale first, and how do I choose the number of components through cross-val? Why could those components that explain substantial predictor variation still be poor predictors of the response?
6. XGBoost and model comparison: How do successive trees improve predictions, and what does reg_alpha penalize? Show how to tune it without using the test set; then compare all models fairly using test mse without overstating the results.